

Arkan Home

E-Commerce Data Localization & Business Intelligence Analytics

An end-to-end analytics project transforming raw e-commerce data into a localized Egyptian business intelligence solution using Excel, SQL Server, Star Schema modeling, DAX, and Power BI.

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EXCEL

DATA CLEANING

DATA VALIDATION

SQL SERVER

ETL

STAR SCHEMA

DAX

POWER BI

DASHBOARD DESIGN

BUSINESS INSIGHTS



Project Executive Summary



From raw e-commerce data to localized business intelligence insights

Arkan Home is an end-to-end analytics project that transforms the public Olist e-commerce dataset into a localized Egyptian business case for a simulated Home Decor and Furniture company.

Business Context

Localized the original e-commerce dataset into an Egyptian market scenario focused on Home Decor and Furniture business performance.

Data Engineering

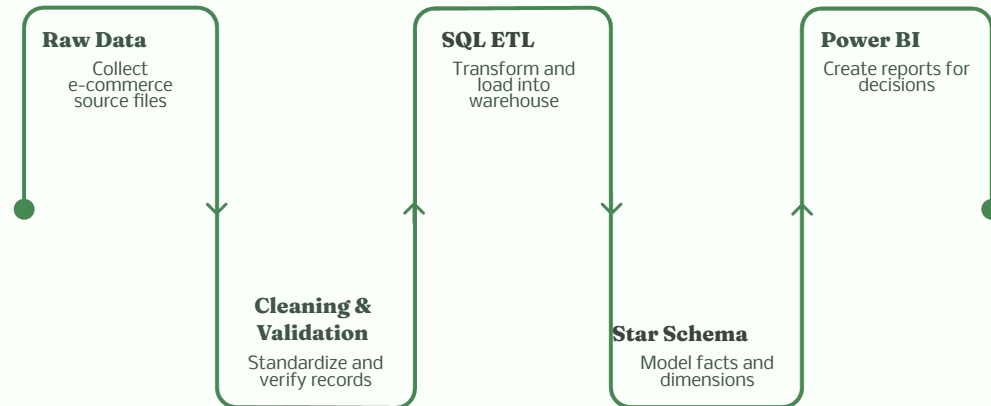
Prepared, cleaned, validated, and transformed raw transactional data using Excel, SQL Server, and structured ETL logic.

Analytical Modeling

Built a Star Schema model with Fact_Sales and dimension tables to support accurate, scalable, and business-ready analytics.

Business Insights

Delivered an interactive Power BI dashboard covering executive performance, sales, logistics, customer behavior, and growth analysis.



Key Message: We did not just build a dashboard. We built a complete business intelligence solution.

Dataset & Localization Strategy



How we transformed the original Olist data into an Egyptian business case

The original Brazilian e-commerce dataset was transformed into a localized Egyptian business scenario to make the analysis more market-relevant and business-oriented.



Source Dataset

Brazilian Olist E-Commerce Dataset. Main source tables: Orders, Order Items, Payments, Reviews, Products, Customers, Sellers.

Localization Assets

- Geography_Mapping.xlsx
- product_category_mapping.csv
- State Flags.csv

Localization Logic

- Mapped original geography to Egyptian cities and regions
- Reclassified product categories into Home Decor & Furniture groups
- Adapted the business scenario to fit a realistic Egyptian e-commerce case



Key Message: We did not just analyze a dataset. We localized it into a market-specific business case.



The SQL layer was built as a controlled data pipeline, starting with raw staging tables, followed by cleaning, validation, ETL transformation, and star schema modeling.

Created the initial raw layer inside SQL Server under **Home_Arkan_DB** to receive the original source entities in a structured staging environment.

```

1  ALTER TABLE slat_order_items (
2      CONSTRAINT slat_order_items_pk PRIMARY KEY (
3          order_id,
4          item_id
5      )
6  );
7
8  CREATE INDEX slat_order_items_order_id_idx ON slat_order_items (
9      order_id
10 );
11
12 CREATE INDEX slat_order_items_item_id_idx ON slat_order_items (
13     item_id
14 );
15
16 CREATE TABLE slat_order_status (
17     order_id INT NOT NULL,
18     power_id INT NOT NULL,
19     seller_id INT NOT NULL,
20     warehouse_id INT NOT NULL,
21     receipt_date DATETIME(3) NOT NULL,
22     CONSTRAINT slat_order_status_pk PRIMARY KEY (
23         order_id,
24         power_id
25     )
26 );
27
28 CREATE INDEX slat_order_status_order_id_idx ON slat_order_status (
29     order_id
30 );
31
32 CREATE INDEX slat_order_status_power_id_idx ON slat_order_status (
33     power_id
34 );
35
36 CREATE INDEX slat_order_status_receipt_date_idx ON slat_order_status (
37     receipt_date
38 );
39
40 CREATE TABLE slat_order_reviews (
41     review_id INT NOT NULL,
42     review_date DATETIME(3) NOT NULL,
43     review_content TEXT NOT NULL,
44     review_rating INT NOT NULL,
45     review_creator_id INT NOT NULL,
46     review_creator_name VARCHAR(255) NOT NULL,
47     review_creator_email VARCHAR(255) NOT NULL,
48     review_creator_phone VARCHAR(255) NOT NULL,
49     review_creator_address VARCHAR(255) NOT NULL,
50     review_creator_city VARCHAR(255) NOT NULL,
51     review_creator_state VARCHAR(255) NOT NULL,
52     review_creator_zip VARCHAR(255) NOT NULL,
53     review_creator_country VARCHAR(255) NOT NULL,
54     review_creator_gender VARCHAR(255) NOT NULL,
55     review_creator_age INT NOT NULL,
56     review_creator_height INT NOT NULL,
57     review_creator_weight INT NOT NULL,
58     review_creator_hair_color VARCHAR(255) NOT NULL,
59     review_creator_eye_color VARCHAR(255) NOT NULL,
60     review_creator_skin_color VARCHAR(255) NOT NULL,
61     review_creator_blood_type VARCHAR(255) NOT NULL,
62     review_creator_marital_status VARCHAR(255) NOT NULL,
63     review_creator_education VARCHAR(255) NOT NULL,
64     review_creator_occupation VARCHAR(255) NOT NULL,
65     review_creator_income INT NOT NULL,
66     review_creator_net_worth INT NOT NULL,
67     review_creator_credits INT NOT NULL,
68     review_creator_debits INT NOT NULL,
69     review_creator_assets INT NOT NULL,
70     review_creator_liabilities INT NOT NULL,
71     review_creator_net_income INT NOT NULL,
72     review_creator_net_worth_2 INT NOT NULL,
73     review_creator_credits_2 INT NOT NULL,
74     review_creator_debits_2 INT NOT NULL,
75     review_creator_assets_2 INT NOT NULL,
76     review_creator_liabilities_2 INT NOT NULL,
77     review_creator_net_income_2 INT NOT NULL,
78     review_creator_net_worth_3 INT NOT NULL,
79     review_creator_credits_3 INT NOT NULL,
80     review_creator_debits_3 INT NOT NULL,
81     review_creator_assets_3 INT NOT NULL,
82     review_creator_liabilities_3 INT NOT NULL,
83     review_creator_net_income_3 INT NOT NULL,
84     review_creator_net_worth_4 INT NOT NULL,
85     review_creator_credits_4 INT NOT NULL,
86     review_creator_debits_4 INT NOT NULL,
87     review_creator_assets_4 INT NOT NULL,
88     review_creator_liabilities_4 INT NOT NULL,
89     review_creator_net_income_4 INT NOT NULL,
90     review_creator_net_worth_5 INT NOT NULL,
91     review_creator_credits_5 INT NOT NULL,
92     review_creator_debits_5 INT NOT NULL,
93     review_creator_assets_5 INT NOT NULL,
94     review_creator_liabilities_5 INT NOT NULL,
95     review_creator_net_income_5 INT NOT NULL,
96     review_creator_net_worth_6 INT NOT NULL,
97     review_creator_credits_6 INT NOT NULL,
98     review_creator_debits_6 INT NOT NULL,
99     review_creator_assets_6 INT NOT NULL,
100    review_creator_liabilities_6 INT NOT NULL,
101    review_creator_net_income_6 INT NOT NULL,
102    review_creator_net_worth_7 INT NOT NULL,
103    review_creator_credits_7 INT NOT NULL,
104    review_creator_debits_7 INT NOT NULL,
105    review_creator_assets_7 INT NOT NULL,
106    review_creator_liabilities_7 INT NOT NULL,
107    review_creator_net_income_7 INT NOT NULL,
108    review_creator_net_worth_8 INT NOT NULL,
109    review_creator_credits_8 INT NOT NULL,
110    review_creator_debits_8 INT NOT NULL,
111    review_creator_assets_8 INT NOT NULL,
112    review_creator_liabilities_8 INT NOT NULL,
113    review_creator_net_income_8 INT NOT NULL,
114    review_creator_net_worth_9 INT NOT NULL,
115    review_creator_credits_9 INT NOT NULL,
116    review_creator_debits_9 INT NOT NULL,
117    review_creator_assets_9 INT NOT NULL,
118    review_creator_liabilities_9 INT NOT NULL,
119    review_creator_net_income_9 INT NOT NULL,
120    review_creator_net_worth_10 INT NOT NULL,
121    review_creator_credits_10 INT NOT NULL,
122    review_creator_debits_10 INT NOT NULL,
123    review_creator_assets_10 INT NOT NULL,
124    review_creator_liabilities_10 INT NOT NULL,
125    review_creator_net_income_10 INT NOT NULL,
126    review_creator_net_worth_11 INT NOT NULL,
127    review_creator_credits_11 INT NOT NULL,
128    review_creator_debits_11 INT NOT NULL,
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131    review_creator_net_income_11 INT NOT NULL,
132    review_creator_net_worth_12 INT NOT NULL,
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134    review_creator_debits_12 INT NOT NULL,
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138    review_creator_net_worth_13 INT NOT NULL,
139    review_creator_credits_13 INT NOT NULL,
140    review_creator_debits_13 INT NOT NULL,
141    review_creator_assets_13 INT NOT NULL,
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144    review_creator_net_worth_14 INT NOT NULL,
145    review_creator_credits_14 INT NOT NULL,
146    review_creator_debits_14 INT NOT NULL,
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148    review_creator_liabilities_14 INT NOT NULL,
149    review_creator_net_income_14 INT NOT NULL,
150    review_creator_net_worth_15 INT NOT NULL,
151    review_creator_credits_15 INT NOT NULL,
152    review_creator_debits_15 INT NOT NULL,
153    review_creator_assets_15 INT NOT NULL,
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155    review_creator_net_income_15 INT NOT NULL,
156    review_creator_net_worth_16 INT NOT NULL,
157    review_creator_credits_16 INT NOT NULL,
158    review_creator_debits_16 INT NOT NULL,
159    review_creator_assets_16 INT NOT NULL,
160    review_creator_liabilities_16 INT NOT NULL,
161    review_creator_net_income_16 INT NOT NULL,
162    review_creator_net_worth_17 INT NOT NULL,
163    review_creator_credits_17 INT NOT NULL,
164    review_creator_debits_17 INT NOT NULL,
165    review_creator_assets_17 INT NOT NULL,
166    review_creator_liabilities_17 INT NOT NULL,
167    review_creator_net_income_17 INT NOT NULL,
168    review_creator_net_worth_18 INT NOT NULL,
169    review_creator_credits_18 INT NOT NULL,
170    review_creator_debits_18 INT NOT NULL,
171    review_creator_assets_18 INT NOT NULL,
172    review_creator_liabilities_18 INT NOT NULL,
173    review_creator_net_income_18 INT NOT NULL,
174    review_creator_net_worth_19 INT NOT NULL,
175    review_creator_credits_19 INT NOT NULL,
176    review_creator_debits_19 INT NOT NULL,
177    review_creator_assets_19 INT NOT NULL,
178    review_creator_liabilities_19 INT NOT NULL,
179    review_creator_net_income_19 INT NOT NULL,
180    review_creator_net_worth_20 INT NOT NULL,
181    review_creator_credits_20 INT NOT NULL,
182    review_creator_debits_20 INT NOT NULL,
183    review_creator_assets_20 INT NOT NULL,
184    review_creator_liabilities_20 INT NOT NULL,
185    review_creator_net_income_20 INT NOT NULL,
186    review_creator_net_worth_21 INT NOT NULL,
187    review_creator_credits_21 INT NOT NULL,
188    review_creator_debits_21 INT NOT NULL,
189    review_creator_assets_21 INT NOT NULL,
190    review_creator_liabilities_21 INT NOT NULL,
191    review_creator_net_income_21 INT NOT NULL,
192    review_creator_net_worth_22 INT NOT NULL,
193    review_creator_credits_22 INT NOT NULL,
194    review_creator_debits_22 INT NOT NULL,
195    review_creator_assets_22 INT NOT NULL,
196    review_creator_liabilities_22 INT NOT NULL,
197    review_creator_net_income_22 INT NOT NULL,
198    review_creator_net_worth_23 INT NOT NULL,
199    review_creator_credits_23 INT NOT NULL,
200    review_creator_debits_23 INT NOT NULL,
201    review_creator_assets_23 INT NOT NULL,
202    review_creator_liabilities_23 INT NOT NULL,
203    review_creator_net_income_23 INT NOT NULL,
204    review_creator_net_worth_24 INT NOT NULL,
205    review_creator_credits_24 INT NOT NULL,
206    review_creator_debits_24 INT NOT NULL,
207    review_creator_assets_24 INT NOT NULL,
208    review_creator_liabilities_24 INT NOT NULL,
209    review_creator_net_income_24 INT NOT NULL,
210    review_creator_net_worth_25 INT NOT NULL,
211    review_creator_credits_25 INT NOT NULL,
212    review_creator_debits_25 INT NOT NULL,
213    review_creator_assets_25 INT NOT NULL,
214    review_creator_liabilities_25 INT NOT NULL,
215    review_creator_net_income_25 INT NOT NULL,
216    review_creator_net_worth_26 INT NOT NULL,
217    review_creator_credits_26 INT NOT NULL,
218    review_creator_debits_26 INT NOT NULL,
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226    review_creator_liabilities_27 INT NOT NULL,
227    review_creator_net_income_27 INT NOT NULL,
228    review_creator_net_worth_28 INT NOT NULL,
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230    review_creator_debits_28 INT NOT NULL,
231    review_creator_assets_28 INT NOT NULL,
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233    review_creator_net_income_28 INT NOT NULL,
234    review_creator_net_worth_29 INT NOT NULL,
235    review_creator_credits_29 INT NOT NULL,
236    review_creator_debits_29 INT NOT NULL,
237    review_creator_assets_29 INT NOT NULL,
238    review_creator_liabilities_29 INT NOT NULL,
239    review_creator_net_income_29 INT NOT NULL,
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241    review_creator_credits_30 INT NOT NULL,
242    review_creator_debits_30 INT NOT NULL,
243    review_creator_assets_30 INT NOT NULL,
244    review_creator_liabilities_30 INT NOT NULL,
245    review_creator_net_income_30 INT NOT NULL,
246    review_creator_net_worth_31 INT NOT NULL,
247    review_creator_credits_31 INT NOT NULL,
248    review_creator_debits_31 INT NOT NULL,
249    review_creator_assets_31 INT NOT NULL,
250    review_creator_liabilities_31 INT NOT NULL,
251    review_creator_net_income_31 INT NOT NULL,
252    review_creator_net_worth_32 INT NOT NULL,
253    review_creator_credits_32 INT NOT NULL,
254    review_creator_debits_32 INT NOT NULL,
255    review_creator_assets_32 INT NOT NULL,
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278    review_creator_debits_36 INT NOT NULL,
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280    review_creator_liabilities_36 INT NOT NULL,
281    review_creator_net_income_36 INT NOT NULL,
282    review_creator_net_worth_37 INT NOT NULL,
283    review_creator_credits_37 INT NOT NULL,
284    review_creator_debits_37 INT NOT NULL,
285    review_creator_assets_37 INT NOT NULL,
```

Replaced blank values, standardized data types, and verified row counts across core transactional and mapping tables.

```

1 ALTER TABLE products ADD (name VARCHAR(255));
2 GO
3
4 UPDATE products SET product_name_length = NULL WHERE product_name_length IS NULL;
5 GO
6 UPDATE products SET product_physics_length = NULL WHERE product_physics_length IS NULL;
7 GO
8 UPDATE products SET product_description_length = NULL WHERE product_description_length IS NULL;
9 GO
10 UPDATE products SET product_length_cm = NULL WHERE product_length_cm IS NULL;
11 GO
12 UPDATE products SET product_width_cm = NULL WHERE product_width_cm IS NULL;
13 GO
14
15 ALTER TABLE products ALTER COLUMN product_name_length INT;
16 ALTER TABLE products ALTER COLUMN product_physics_length INT;
17 ALTER TABLE products ALTER COLUMN product_description_length INT;
18 ALTER TABLE products ALTER COLUMN product_length_cm INT;
19 ALTER TABLE products ALTER COLUMN product_width_cm INT;
20 GO
21
22 SELECT * FROM Table_Name_CNAME() AS Row_Count FROM products;
23 GO
24
25 SELECT * FROM CNAME() FROM vider_order;
26 GO
27
28 SELECT * FROM customers; CNAME() FROM vider_order;
29 GO
30
31 SELECT * FROM sellers; CNAME() FROM vider_order;
32 GO
33
34 SELECT * FROM products; CNAME() FROM vider_order;
35 GO
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37 SELECT * FROM customers; CNAME() FROM vider_order;
38 GO
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40 SELECT * FROM sellers; CNAME() FROM vider_order;
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343 SELECT * FROM customers; CNAME() FROM vider_order;
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346 SELECT * FROM sellers; CNAME() FROM vider_order;
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349 SELECT * FROM products; CNAME() FROM vider_order;
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352 SELECT * FROM customers; CNAME() FROM vider_order;
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478 SELECT * FROM customers; CNAME() FROM vider_order;
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```

Built mapping tables, dimension tables, and the central **Fact_Sales** table to prepare the data for scalable Power BI reporting.

```

1  #!/usr/bin/perl -w
2  use strict;
3  use Data::Dumper;
4  use DBI;
5  use DBD::Oracle;
6  use DBD::Pg;
7  use DBD::SQLite;
8  use DBD::Sybase;
9  use DBD::Interbase;
10 use DBD::Informatica;
11 use DBD::Microsoft;
12 use DBD::MySQL;
13 use DBD::OCI;
14 use DBD::ODBC;
15 use DBD::PostgreSQL;
16 use DBD::SQLanywhere;
17 use DBD::Teradata;
18 use DBD::Vista;
19 use DBD::WebDB;
20 use DBD::XDBC;
21 use DBD::XDBC2;
22 use DBD::XDBC3;
23 use DBD::XDBC4;
24 use DBD::XDBC5;
25 use DBD::XDBC6;
26 use DBD::XDBC7;
27 use DBD::XDBC8;
28 use DBD::XDBC9;
29 use DBD::XDBC10;
30 use DBD::XDBC11;
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311 use DBD::XDBC292;
312 use DBD::XDBC293;
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 **Key Message:** The SQL pipeline transformed raw data into a trusted analytical foundation. This layer prepared the project for accurate modeling, DAX measures, and Power BI reporting.

Power BI Dashboard Architecture



Connecting model design, navigation, and interactive business reporting

The Power BI solution was designed as a connected analytical experience, combining a branded landing page, structured data model, interactive navigation, slicers, and dashboard pages.



Dashboard Ecosystem

- Landing Page
- Executive Overview
- Sales
- Logistics
- Customer Behavior
- Growth Analysis

Model Foundation

- Central Fact_Sales table
- Connected dimension tables
- Consistent relationship logic
- Clean filtering and scalable reporting

Interactivity & UX

- Page navigation
- Year and month slicers
- Cross-filtering across visuals
- Help overlay and guided user flow



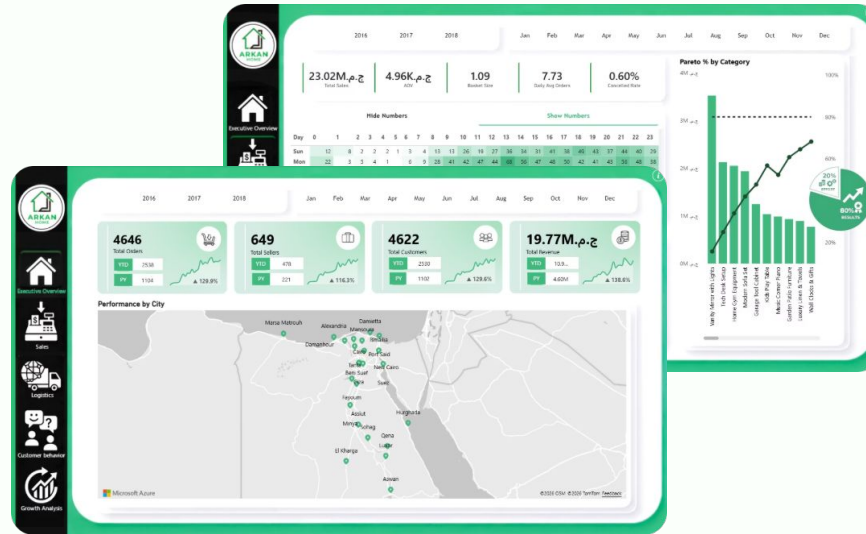
Key Message: The Power BI layer converted the analytical model into a guided business intelligence experience. It connected navigation, visuals, and business domains into one decision-support system.

Executive & Sales Insights



Combining leadership overview with commercial performance analysis

The executive and sales dashboards provide a clear view of business scale, revenue performance, order behavior, category contribution, payment methods, and market distribution.



Key Message: Executive and sales analytics translate transactions into commercial intelligence. They help stakeholders understand performance, revenue drivers, and market activity.

Logistics & Customer Behavior Insights



Connecting operational performance with customer experience

The logistics and customer behavior dashboards help connect fulfillment performance, delivery reliability, acquisition patterns, retention signals, review scores, and customer satisfaction.



Logistics Performance

- Avg Dispatch Days: **3.17**
- Late Dispatch %: **8.99%**
- Avg Delivery Days: **12.16**
- Late Delivery %: **8.12%**
- Late Deliveries: **368**

Customer Behavior

- Total Customers: **4,622**
- Returning Customers: **24**
- Orders per Customer: **1.01**
- Avg Customer Spend: **4.98K**
- Avg CSAT Score: **4.08**

Decision Support Value

- Identifies logistics bottlenecks
- Highlights customer satisfaction signals
- Connects delivery performance with customer experience
- Supports service improvement actions



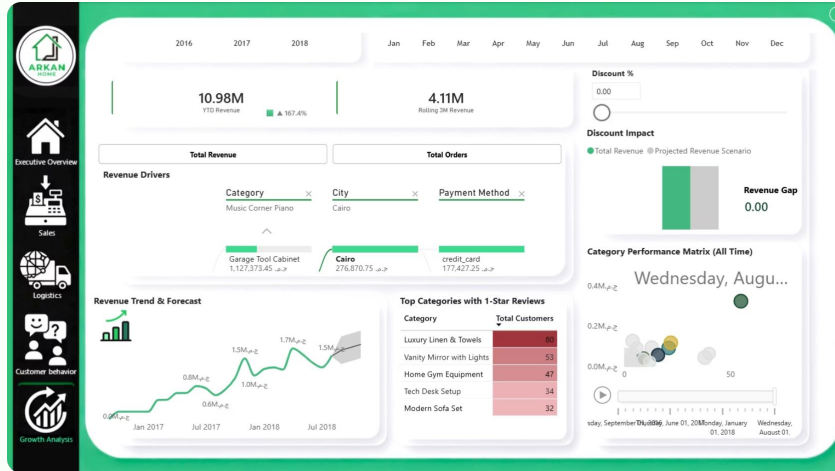
Key Message: Operational intelligence becomes stronger when logistics and customer behavior are analyzed together. This view connects delivery performance with satisfaction, retention, and service improvement opportunities.

Growth Analysis & Decision Support



Turning business performance into strategic planning insights

The growth analysis dashboard supports strategic review by combining revenue trends, category contribution, discount scenario analysis, forecast movement, and performance risk indicators.



Growth Drivers

- Revenue driver decomposition
- Category, city, and payment method impact
- Top categories with weak review signals

Scenario Thinking

- Discount impact simulation
- Projected revenue comparison
- Revenue gap visibility
- Revenue trend and forecast review

Decision Support

- Supports growth planning
- Identifies revenue leakage areas
- Reveals category-level risks
- Connects performance with action priorities



Key Message: Growth analysis turns performance history into future-focused decision support. It helps compare revenue drivers, scenarios, and category risks in one analytical view.

Technical Skills, DAX & Challenges



The technical foundation behind the final business intelligence solution

The project demonstrates a full BI workflow across data review, SQL engineering, analytical modeling, DAX logic, dashboard design, and business analysis.

Skills Applied

Excel & Data Review

SQL Server

Data Cleaning

ETL

Star Schema

DAX

Power BI

Business Analysis

DAX & KPI Logic

- Total Revenue
- AOV
- Basket Size
- Daily Avg Orders
- Late Delivery %
- Returning Customers
- Avg CSAT Score
- Projected Revenue Scenario

Challenges Solved

- Data quality inconsistencies
- Localization of a non-Egyptian dataset
- Modeling raw transactions into star schema
- Designing a usable multi-page dashboard experience



Key Message: The project proves capability across the full BI lifecycle. Technical execution and business analysis were combined into one complete decision-support solution.

Final Outcomes & Closing



From raw data to localized business intelligence

Arkan Home transformed raw e-commerce data into a complete localized BI solution that supports executive review, sales analysis, logistics monitoring, customer understanding, growth planning, and decision support.

End-to-End BI Solution

From raw data preparation to final dashboard storytelling.

Localized Business Case

Converted the original dataset into an Egyptian Home Decor and Furniture scenario.

Decision-Support Dashboard

Delivered interactive pages for executive, sales, logistics, customer, and growth analysis.

Team: Ahmed Magdy · Mark Morris · Mai Ibrahim · Aisha Taha

Thank You



From raw data to business intelligence. A complete analytics journey designed to support smarter decisions.